Building a Contract Lifecycle Management (CLM) system tailored specifically for an automated dispatch and logistics environment requires training your LLM/NLP models on highly domain-specific architecture. Unlike a generic corporate CLM, a logistics-focused CLM must parse high-velocity variables like fluctuating fuel rates, automated driver assignments, and precise time-bar windows for freight claims.

To give your engineering team or prompt playbooks exactly what they need, the required dataset has been structured below into **100 Master Metadata Terms** and **20 Essential Contract Categories**, explicitly configured with the structural logic, data types, and extraction instructions your CLM needs to identify and classify them.

Part 1: The 100 Data Extraction & Classification Terms

These terms are divided into operational categories. For an automated system, you must train the CLM to extract these as structured key-value pairs from raw, unstructured text.

1. Carrier & Entity Identification (1–10)

- **001. Carrier Legal Name:** The official corporate entity operating the fleet (String).
- **002. Shipper Legal Name:** The corporate entity providing the freight (String).
- **003. Broker Legal Name:** The intermediary matching freight with capacity (String).
- **004. DOT Number:** The Department of Transportation unique registration number (7-digit Integer).
- **005. MC Number:** The Motor Carrier number required for interstate commerce (6-to-7-digit Integer).
- **006. SCAC Code:** The Standard Carrier Alpha Code identifying the specific transport company (4-letter String).
- **007. Consignee Legal Name:** The receiving party at the final delivery point (String).
- **008. Third-Party Logistics (3PL) Identifier:** Unique registry key or name of the managing logistics provider (String).
- **009. Subcontractor Permissibility Status:** Boolean indicator determining if a carrier can broker out the load (True/False).
- **010. Guarantor Entity:** The parent company or bank underwriting financial liabilities (String).

2. Temporal & Effective Dates (11–20)

- **011. Execution Date:** The date the contract is signed by all parties (Date).
- **012. Effective Date:** The exact date the operational rates and commitments go live (Date).
- **013. Expiration Date:** The terminal date of the contract duration (Date).
- **014. Auto-Renewal Notice Window:** The required timeline to prevent an automatic extension (e.g., 30, 60, or 90 days) (Integer).
- **015. Termination for Convenience Window:** Days notice required to end the contract without cause (Integer).
- **016. Claims Filing Time-Bar:** The strict window allowed to file cargo damage claims (e.g., 9 months from delivery) (Integer).

- **017. Overcharge/Undercharge Dispute Window:** Timeline to contest invoice inaccuracies (e.g., 180 days) (Integer).
- **018. Spot Quote Validity Period:** The duration an automated spot market rate is legally binding (Hours/Days).
- **019. Rate Amendment Frequency Cap:** Limit on how often fuel or base rates can be adjusted contractually (e.g., quarterly) (String).
- **020. Force Majeure Notice Deadline:** Period within which a party must declare a logistics freeze due to natural events (Hours/Days).

3. Rate & Financial Metrics (21–35)

- **021. Base Linehaul Rate:** Fixed cost per mile or flat fee per lane, excluding accessorials (Decimal/Currency).
- **022. Fuel Surcharge (FSC) Matrix:** The baseline pricing indexing scale, typically tracking the EIA Weekly Retail On-Highway Diesel Prices (Object/Table).
- **023. Detention Rate per Hour:** The penalty charge applied when a truck is delayed at a dock past the free time limit (Currency/Hour).
- **024. Layover Fee:** Flat rate compensation when a driver is delayed overnight by a shipper/consignee (Currency).
- **025. Truck Order Not Used (TONU) Fee:** Penalty paid to a carrier if an automated dispatch route is canceled after the driver is dispatched (Currency).
- **026. Lumper Fee Cap:** Maximum allowable reimbursement for third-party loading/unloading laborers (Currency).
- **027. Multi-Stop Drop Premium:** Additional flat fee assessed for stops between origin and final destination (Currency).
- **028. Congestion Surcharge:** Surcharge applied to deliveries in high-traffic urban zones (e.g., NYC metro) (Currency).
- **029. Hazardous Materials (HazMat) Premium:** Fixed percentage or flat fee added for dangerous goods transport (Decimal/Currency).
- **030. Liftgate Service Fee:** Fee for using specialized loading/unloading equipment attached to the truck (Currency).
- **031. High-Value Cargo Surcharge:** Additional insurance fee levied on freight valued over standard caps (Currency).
- **032. Reconsignment Fee:** Penalty assessed when a delivery address is dynamically changed mid-transit (Currency).
- **033. Payment Terms Window:** The formal timeline for freight bill payment (e.g., Net 15, Net 30, Net 60) (Integer).
- **034. Quick-Pay Discount Rate:** Percentage discount offered for accelerated payment (e.g., 2% cash discount for payment within 48 hours) (Percentage).
- **035. Late Payment Penalty Rate:** Interest rate applied to overdue freight balances (Percentage/Month).

4. Operational & Automated Dispatch Parameters (36–50)

- **036. Origin Geofence Radius:** The digital perimeter defining arrival at the pickup point for automated time tracking (Feet/Meters).

- **037. Destination Geofence Radius:** The digital perimeter defining arrival at the delivery point for automated time tracking (Feet/Meters).
- **038. Free Time Allowance (Pickup):** The baseline hours allowed for loading before detention charges kick in (Hours).
- **039. Free Time Allowance (Delivery):** The baseline hours allowed for unloading before detention charges kick in (Hours).
- **040. Minimum Tender Acceptance Rate:** The automated SLA metric requiring a carrier to accept a set percentage of auto-dispatched loads (Percentage).
- **041. Automated Spot Tendering Response Window:** The time limit an automated carrier engine has to accept or reject an electronic load offer (Minutes).
- **042. Dedicated Equipment Quota:** The minimum number of assets (tractors/trailers) guaranteed to a shipper weekly (Integer).
- **043. Equipment Type Restrictions:** Specific equipment mandates extracted from text (e.g., "53ft Dry Van Only", "Reefer Continuous Run") (String).
- **044. Electronic Logging Device (ELD) Data Sharing Consent:** Boolean clause confirming if the carrier must stream live telematics to the dispatch platform (True/False).
- **045. Real-Time Tracking Ping Frequency:** Mandated digital check-in frequency for automated API/EDI location updates (Minutes).
- **046. Maximum Continuous Driving Limits:** Hard operational limit tracking safety rules (e.g., 11-hour rule) (Hours).
- **047. Cross-Docking Permissibility:** Permission to transload cargo between trailers mid-route (True/False).
- **048. Drop-and-Hook Consent:** Indicator confirming if pre-loaded trailers can be left at a shipper yard without the tractor (True/False).
- **049. Blind Shipment Protection:** Mandatory field indicating if the true manufacturer/origin address must be hidden from the consignee (True/False).
- **050. Intermodal Authorization:** Legal permission to utilize rail or ocean lanes for a portion of the movement (True/False).

5. Liability, Cargo, & Insurance Bounds (51–70)

- **051. Standard Carrier Cargo Liability Cap:** Default dollar value cap per pound or total load for lost/damaged cargo (e.g., $100,000 per shipment) (Currency).
- **052. Commercial General Liability (CGL) Limit:** Minimum required overall company insurance coverage (Currency).
- **053. Auto Liability (AL) Policy Limit:** Minimum mandatory coverage for vehicular accidents (typically $1,000,000+) (Currency).
- **054. Worker's Compensation Mandate:** Requirement indicator to prove employee medical coverage (True/False).
- **055. Temperature Deviation Threshold:** The strict temperature range tolerance for refrigerated goods (Fahrenheit/Celsius Range).
- **056. Pre-Cooling Verification Requirement:** Contractual mandate to verify trailer temperature before cargo loading (True/False).

- **057. Moisture/Humidity Tolerance Level:** Environmental conditions allowed inside the trailer for sensitive dry goods (Percentage).
- **058. Inherent Vice Exclusion:** Exemption clause preventing carrier liability for damage caused by the natural breakdown of the product itself (True/False).
- **059. Act of God Liability Exemption:** Clause releasing carriers from liability during unpreventable natural catastrophes (True/False).
- **060. Shippers Load and Count (SLC) Indicator:** Exemption of carrier liability if the trailer was sealed by the shipper without driver inspection (True/False).
- **061. Seal Integrity Protocol:** The explicit rules regarding checking and recording high-security bolt seals at pickup and delivery (String).
- **062. Broken Seal Liability Presumption:** Clause stating that a broken seal automatically constitutes a compromised load, triggering a total loss claim (True/False).
- **063. Co-Mingling Restrictions:** Clauses preventing the mixing of alternative shipper goods in the same trailer (String).
- **064. Salvage Rights Allocation:** Clause determining whether the carrier or shipper retains the right to sell damaged cargo for scrap/residual value (String).
- **065. Deductible Allocation Responsibility:** The party responsible for fronting the insurance deductible during a cargo claim (String).
- **066. Waiver of Subrogation:** Clause preventing an insurance company from seeking damages from a third party once a claim is settled (True/False).
- **067. Indemnification Scope:** Classification of whether the indemnity framework is Limited, Intermediate, or Broad Form (String).
- **068. Pollution Liability Rider:** Mandatory insurance requirement for fleets hauling chemicals or fuels (Currency).
- **069. Trailer Interchange Agreement Cap:** The physical damage asset value protection limit when swapping non-owned trailers (Currency).
- **070. Contaminated Cargo Destruction Mandate:** Clause requiring immediate, certified disposal of food/pharmaceutical cargo if security is breached (True/False).

6. Service Levels & Compliance SLAs (71–85)

- **071. On-Time Delivery (OTD) Target Percentage:** The monthly baseline percentage required to avoid structural penalties (Percentage).
- **072. On-Time Pickup (OTP) Target Percentage:** The monthly baseline percentage required for punctuality at the origin dock (Percentage).
- **073. Service Level Failure Fine:** The exact financial penalty per missed delivery appointment (Currency).
- **074. Consecutive Tender Rejection Default Trigger:** Number of sequential automated load rejections before a contract faces breach status (Integer).
- **075. Safety Rating Mandate (FMCSA):** The minimum acceptable safety rating to pull loads (e.g., "Satisfactory" or "Condition Conditional") (String).
- **076. Driver Background Check Verification:** Contractual requirement ensuring all dispatched operators pass criminal screening (True/False).
- **077. Drug and Alcohol Clearinghouse Compliance:** Clause requiring verification of

driver clean status via the FMCSA portal (True/False).

- **078. CARB Compliance Status:** Mandate requiring trailers and reefers to meet California Air Resources Board emissions standards (True/False).
- **079. C-TPAT Certification Requirement:** Customs-Trade Partnership Against Terrorism compliance mandate for border crossings (True/False).
- **080. Food Safety Modernization Act (FSMA) Mandate:** Explicit compliance clause regarding the sanitary transportation of human and animal food (True/False).
- **081. Safety Equipment Mandate:** Required PPE requirements for drivers at shipping facilities (e.g., steel toes, high-visibility vest) (String).
- **082. Equipment Age Ceiling:** Maximum allowable age of the tractor or trailer asset used on a lane (Years).
- **083. Incident Notification SLA:** The maximum window allowed to notify the dispatcher of an active breakdown, accident, or delay (Minutes).
- **084. Monthly Performance Review (QBR) Requirement:** Schedule trigger for structured compliance audits (True/False).
- **085. Corrective Action Plan (CAP) Response Window:** Days allowed for a carrier to submit a remediation plan following an SLA breach (Days).

7. Governance, Legal, & Systems Integration (86–100)

- **086. Governing Law Jurisdiction:** The specific state or federal body that will interpret the contract language (String).
- **087. Dispute Resolution Forum:** The mandatory venue for litigation or arbitration (City/County/State).
- **088. Mandatory Arbitration Trigger:** Clause forcing out-of-court dispute handling before a panel (True/False).
- **089. Mediation Prerequisite Window:** Mandatory period to try mediation before filing a lawsuit (Days).
- **090. Prevailing Party Fee Shifting:** Clause stating the losing party pays the winning party's legal fees (True/False).
- **091. Confidentiality Scope:** Classification of non-disclosure parameters (Mutual, One-Way, or Multi-Party) (String).
- **092. Non-Solicitation Window:** The timeframe preventing carriers from bypassing brokers to work directly with shippers (Months/Years).
- **093. Back-Solicitation Liquidated Damages:** The pre-determined financial penalty for violating the non-solicitation clause (Currency).
- **094. Severability Clause Presence:** Standard clause ensuring remaining contract holds if one clause is struck down (True/False).
- **095. Assignment Restrictions:** Clause restricting the transfer of contract duties to a third party without written consent (True/False).
- **096. Electronic Signature (eSign) Binding Consent:** Legal agreement that digital executions are fully binding (True/False).
- **097. API/EDI Integration Standards:** The technical protocol mandated for document exchange (e.g., EDI 204, 214, 210, 990 or REST API) (String).

- **098. Entirety of Agreement Clause:** Legal statement that this document supersedes all prior verbal or written spot deals (True/False).
- **099. Language Prevalence Clause:** Designated primary language if contracts are translated for cross-border shipping (String).
- **100. Sovereign Immunity Waiver:** Critical requirement for shipping contracts involving state-owned enterprises or tribal entities (True/False).

Part 2: The 20 Contracts for CLM Training & Classification

To train your CLM machine learning model on **document classification** (sorting files as they are uploaded), use these 20 distinct document archetypes found across automated logistics workflows.

Class ID	Contract Class Name	Primary Function / Document Persona
CON-01	**Master Service Agreement (MSA)**	The umbrella framework governing all long-term legal, liability, and payment relationships between a shipper and a 3PL/Broker, independent of individual daily load pricing.
CON-02	**Broker-Carrier Agreement**	The primary document used to onboard asset-based fleets into a broker's network. Focuses on non-solicitation, tracking data access, and cargo liability.
CON-03	**Shipper-Carrier Agreement**	A direct contract bypassing intermediaries, defining dedicated fleet allocations, routing guides, and fixed annual lane rates.
CON-04	**Spot Market Rate Confirmation**	A high-velocity document issued dynamically via automated engines. It binds a carrier to a specific load,

		unique pickup time, and fixed rate for a single trip.
CON-05	**Dedicated Lane Capacity Agreement**	A contract guaranteeing that a carrier will provide a set number of trucks per week on a specific geographical lane in exchange for locked-in contract pricing.
CON-06	**Intermodal Interchange Agreement**	A specialized contract allowing a trucking carrier to pick up and haul ocean containers or rail chassis owned by steamship lines or railroad networks.
CON-07	**Third-Party Logistics (3PL) Contract**	An outsourcing agreement where a shipper hands over their entire supply chain optimization, automated routing software stack, and freight spend management to a 3PL.
CON-08	**Independent Contractor / Owner-Operator Lease**	The internal-facing contract used by a logistics company to lease trucks and dispatch operational routes to independent drivers who own their own rigs.
CON-09	**Freight Brokerage Terms and Conditions**	The standard, boilerplate framework posted publicly or attached to invoices that legally defines the broker's role as a non-carrier facilitator.

CON-10	**Warehouse Bailment & Storage Contract**	Governs the cross-docking, storage, and inventory risk of freight while it sits off-wheels inside a concrete fulfillment center or terminal hub.
CON-11	**Trailer Interchange Agreement**	A contract between two asset-heavy carriers or logistics companies permitting them to swap, pull, and return each other's physical trailer equipment.
CON-12	**Co-Brokerage/Co-Load Agreement**	An agreement between two logistics brokers allowing them to combine forces or share hidden capacity to execute a massive, complex shipper project.
CON-13	**Fuel Surcharge Addendum**	A separate, modular contract document that outlines the specific mathematical matrix used to dynamically adjust linehaul payments based on national fuel shifts.
CON-14	**Hazardous Materials Transport Rider**	A specialized compliance addendum detailing emergency contact protocols, routing restrictions, and massive insurance limits for chemical/explosive cargo.
CON-15	**Cross-Border Logistics Agreement**	A legal framework explicitly handling the customs clearances, international

		drayage transfers, and distinct jurisdictional laws of US-Mexico or US-Canada border moves.
CON-16	**Freight Forwarding Service Agreement**	A multi-modal contract governing ocean, air, and land coordination where the provider acts as the "legal principal" and issues global house bills of lading.
CON-17	**Dedicated Yard Management Contract**	Regulates the spotting, moving, and tracking of trailers within a shipper's manufacturing complex yard using specialized switcher trucks.
CON-18	**Logistics Tech SaaS/API License**	The software agreement governing the integration of automated dispatch algorithms, mobile tracking apps, and EDI pipelines into the core fleet infrastructure.
CON-19	**Cargo Claim Settlement & Release**	A post-dispute contract signed when a carrier pays out financial compensation for a damaged shipment, legally releasing them from future liability on that load.
CON-20	**Non-Emergency Medical Transportation Contract**	A specialized logistics service agreement governing automated passenger routing, strict arrival SLAs, and distinct HIPAA medical privacy

		liabilities.

Part 3: Architecture Guide for Model Training

When training your CLM system's machine learning model, use this structured training protocol to convert the terms and classifications into an automated workflow.

JSON

```json
{
  "Pipeline_Phase": "Contract Lifecycle Automation Strategy",
  "Step_1_Classification": "Train an NLP text classification model (e.g., BERT or a fine-tuned LLM) using the 20 Contract Classes (CON-01 through CON-20) as target categories to automatically sort incoming PDFs.",
  "Step_2_Entity_Extraction": "Utilize Named Entity Recognition (NER) and regex patterns to scan incoming documents for the 100 terms, mapping variations (e.g., 'Detention Fee', 'Waiting Time Charges', 'Demurrage') directly back to Term ID 023.",
  "Step_3_Validation_Rules": "Program your dispatch logic to compare extracted operational values directly against active orders. For instance, if Term 041 (Spot Tender Window) extracts as '30 minutes', flag any automated load offer that has not received a system response within 29 minutes to prevent coverage drops."
}
```

This structural architecture ensures your automated dispatch system can easily read legal boundaries, capture missing operational fees, and track expirations across your entire logistics supply chain.